崑山科技大學日間部資訊工程系四技課程科目表

Kun Shan University Day Program Department of Information Engineering

Four-Year Technical Program Course Schedule

課程名稱(講授時數－實習時數－學分數)
Course Title (Lecture Hours – Lab Hours – Credits)

114.04.11 系教學與課程規劃委員會議修訂通過
114.04.22 院課程委員會議通過
114.07.22 校課程委員會議通過
113 學年度入學適用
Effective for students admitted in Fall, 2024

第一學年 1st Academic Year		第二學年 2nd Academic Year		第三學年 3rd Academic Year		第四學年 4th Academic Year	
第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester
共同必修科目(合計 29 學分) University-wide Required Courses (29 Credits)							
校訂共同科目(合計 2 學分) General Courses (2 Credits)							
	生活實踐教育(一) Living Practice (I) (0-1-1)	生活實踐教育(二) Living Practice (II) (0-1-1)	服務學習與公民教育 Public Service and Civil Education (0-1-0)				
核心通識科目(合計 17 學分) Common Core Courses (17 Credits)							
體育（一） Physical Education (I) (2-0-2)	體育（二） Physical Education (I) (2-0-2)			生涯發展與職業倫理 Career Development and Work Ethics (工民商 2-0-2)	外語選讀 Foreign Languages ** (工商 2-0-2)		英文能力檢定* English Proficiency Examination* (1-0-1)
中文閱讀與表達(一) Chinese Reading and Expression(I) (2-0-2)	中文閱讀與表達(二) Chinese Reading and Expression(II) (2-0-2)						
英文聽力與閱讀(一) English Listening and Reading (I) (2-0-2)	英文聽力與閱讀(二) English Listening and Reading (II) (2-0-2)						
分類通識科目(合計 10 學分) Liberal Education Courses (10 Credits)							
分類通識 Liberal Education (2-0-2)	分類通識 Liberal Education (2-0-2)	分類通識 Liberal Education (2-0-2)	分類通識 Liberal Education (2-0-2)	分類通識 Liberal Education (2-0-2)			
8-0-8	8-1-9	2-1-3	2-1-2	4-0-4	2-0-2		(1-0-1)

註：Note:

*「英文能力檢定」依「崑山科技大學學生英文基本能力檢定實施辦法」辦理。
'English Proficiency Examination' is implemented according to 'Kun Shan University Guidelines for English Proficiency Requirement.'

**「外語選讀」包括「第二外語」及「職場英文」為選項式必修科目，課程擇一門修習即可。課程開設於三年級，然「第二外語」課程為二年級優先選課。
'Foreign Languages' comprises 'Foreign Languages (other than English)' and 'Workplace English', from which students choose one course. While the courses are offered in the third academic year, 'Foreign Languages (other than English)' is given preference to sophomores.

院訂必修科目（合計 17 學分）**College-Designated Required Courses (Total: 17 Credits)**

第一學年 1st Academic Year		第二學年 2nd Academic Year		第三學年 3rd Academic Year		第四學年 4th Academic Year	
第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester
工程技術概論 Introduction to Engineering Technology (2-0-2)	資訊安全概論 Introduction to Information Security (3-0-3) HTML 5 程式設計 HTML 5 Programming (1-2-3) 科技英文 Technical English (2-0-2)	App 程式設計 App Programming (1-2-3)			學術倫理 Academic Ethics (0-1-1) 電競周邊商品開發 E-sports peripheral product development (1-2-3)		
2-0-2	6-2-8	1-2-3			1-3-4		

註*：「學術倫理」之實施與學分取得依課務組網頁之「學術倫理教育實施要點」辦理。

Note*：The implementation of 'Academic Ethics' and the earning of credits are conducted in accordance with the 'Guidelines for Academic Ethics Education' published on the Curriculum Division's website.

院訂專業模組必修/選修科目（合計 20 學分）Professional Modules Necessary Courses of College (Total： 20 Credits)

智慧生活專業模組課程（資工、電遊）

Smart Living Professional Module Courses

Premiere 影片剪輯 Premiere Movie Clips (1-2-3) (電遊) 資訊技術導論 Introduction to Information technology (3-0-3)	C#程式設計入門 Introduction to C# Programming (2-1-3) (電遊)	創新智慧生活科技 Innovative Smart Life Technology (2-1-3)	行動遊戲製作 Mobile Game Creation (2-1-3) (電遊)	元宇宙概論 Introduction to Metaverse (2-0-2) (電遊)		人工智慧 Artificial Intelligent (3-0-3)	
4-2-6	2-1-3	2-1-3	2-1-3	2-0-2		3-0-3	

專業必修科目(合計 37 學分) Professional Required Courses (Total: 37 Credits)

程式設計 Programming Design （1-2-3）		資料結構 Data Structure (3-0-3)			離散數學 Discrete Mathematics (3-0-3)		程式設計 Programming Design
微積分 Calculus (3-0-3)	工程數學 Engineering Mathematics (3-0-3)			軟體工程 Software Engineering (3-0-3)			軟體開發 Software Development
		*Linux 作業系統實務 Practice of Linux Operating System (1-2-3)	計算機網路實務 Computer Network Practice (1-2-3)	人工智慧與雲端運算概論 Introduction to Artificial Intelligence and Cloud Computing (3-0-3)			雲端運算與網路技術 Cloud Computing and Network Technology
			資料庫實務 practice of database (1-2-3)	嵌入式系統與實習 Course outline of Embedded System and Lab (1-2-3)			嵌入式基礎 Embedded Fundamental
					*專題製作 Special Project （0-2-2）	暑期校外實習 Summer School Internship (0-*-2)	專業認證 Professional certification (0-1-0)	整合性課程 Integrated Courses

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4-2-6	3-0-3	4-2-6	2-4-6	7-2-9	3-2-5	0-0-2	0-1-0	
專業選修科目(依每學期實際情況開課，含專業養成課程 8 學分) Professional Elective Courses (Offered based on the actual circumstances of each semester, including 8 credits of professional development courses)								
數位邏輯與實習 Digital Logic and Practice (2-1-3)		影像處理概論 Introduction to Image Processing (3-0-3)	演算法 Algorithm (3-0-3) 伺服器架設實務 Server Construction Practice (1-2-3)	計算機結構 Computer Architecture (3-0-3)		專案管理 Project Management (3-0-3)	電子商務實務 Practice of E-Commerce (3-0-3)	計算機數學與理論 Mathematics for Computing and Theory
	物聯網於綠能之應用 IOT application in Green Energy (1-2-3)			智慧能源與物聯網 Intelligent Energy and IOT (1-2-3)		物聯網實務 Practice of Internet of Things (1-2-3)	多媒體智慧聯網實務 Multimedia Smart Networking Practice (2-1-3)	智慧聯網 Smart Network
資訊工程導論 Outline of Information Technology Concepts (3-0-3)				車用晶片模組應用系統概論 Introduction to Automotive Chip Module Application Systems (3-0-3)		大數據分析 Big Data Analysis (1-2-3)	雲端服務導向架構中介軟體 Cloud Service-Oriented Architecture Middleware (3-0-3)	智慧服務 Intellegient Service
資訊安全實務 Information Security (1-2-3)	網路攻防實務 Cyber Offense and Defense Practice (1-0-1) CTF 搶旗大賽實務技術 CTF Flag-Raising Contest Practical Technology (1-0-1)	網站安全 Web Security (2-1-3)	系統安全 System security (3-0-3)	雲端安全 Cloud Security (3-0-3)			安全實務 Safety practice (2-1-3) 物聯網與資訊安全 Internet of Things and Information Security (2-1-3)	資訊安全 Information Security
基本電學與實習 Fundamental Circuit and the Practice （1-2-3）	可程式邏輯與實習 Programmable Logic and Practice （1-2-3）	電子電路與實習 Electronics and the Practice (1-2-3)	微算機原理與實習 Microcomputer principle and practices (1-2-3)	嵌入式微處理器系統 Embedded Microprocessor Systems (1-2-3) 電腦視覺實務 Computer Vision Practice (1-2-3)	影像處理 Image Processing (3-0-3) AOI 檢測技術 實務 AOI testing technology practice (1-2-3) 機器學習 Machine Learning (3-0-3)	資電整合實務 Integration of Information and Electrical Engineering (1-2-3) 深度學習實務 Deep Learning Practice (3-0-3)	機器人程式設計 Robotic Programming Design (1-2-3) 智慧聯網自動光學檢測實務專題 Smart Networked AOI Special Project (1-2-3)	智慧應用 Intelligent Application
		Swift IOS app 開發 Swift iOS Applications Development (3-0-3)	行動裝置 App 專題製作 Mobile Device App Project Creation (3-0-3)	iOS 應用開發 iOS app Development (3-0-3)	行動裝置遊戲開發 Mobile Device Game Development (3-0-3)	行動 APP 專案開發 Mobile APP Project Development (3-0-3)		App 開發 App Development
跨網頁平台設計 Cross-web platform design (3-0-3)	網站開發與安全 web development and application security (3-0-3)				D3.js 程式設計 D3.js programming (3-0-3)	JSP 程式設計 JSP programming (3-0-3)	Ajax 程式設計 Ajax Programming (3-0-3)	網站程式設計 Web Programming

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python 程式設計 python Programming （3-0-3）				伺服網頁程式設計 Server Web Design (1-2-3)			設計模式實務 Design Pattern Practice (1-1-2)	程式設計 Programming Design
視覺設計實務 Visual Design Practice (3-0-3)	網頁視覺設計實務 Web Visual Design Practice (3-0-3)			手機 UI 設計 Mobile UI Design (3-0-3)		資料視覺化 Data Visualization (3-0-3)		互動設計 Interactive Design
				專業校外實習(一) Industrial Internship(I) (0-2-2)	專業校外實習(二) Industrial Internship(II) (0-2-2)	學期校外實習(一) Semester Internship (I) (0-*-9) 學年校外實習(一) Annual Internship (I) (0-*-9)	學期校外實習(二) Semester Internship (II) (0-*-9) 學年校外實習(二) Annual Internship (II) (0-*-9)	企業實習 Industrial Internship
微服務伺服器實務 Microservice server practice (0-1-1) 專案開發與應用實務 Project Developement and Application Practice (0-1-1) 關聯式資料庫實務 Relational Database Practice (0-1-1)	GTM 實務 Google Tag Manager Practice (0-1-1) 軟體測試實務 Software Testing Practicalities (0-1-1) 人機介面設計專題 Human Machine Interface Design Project (0-1-1)	資電整合應用 Applications of Information and electronic integration (0-1-1) 跨平台程式設計專題 Cross-platform programming languages (0-1-1) 網頁前後端整合實務 Web Integration Practices on Front-End and Back-End (0-1-1)	M2M 聯網實務 Practice of Machine to Machine (0-1-1) 爬蟲程式應用實務 Application of python in data crawling (0-1-1) 物聯網巨量資料實務 IOT BigData Practice (0-1-1)	微電腦視覺實務 Microcomputer Vision and Practice (0-1-1) 網路串流技術應用 The Application of Network Streaming Technology (0-1-1) 機器學習實務 Machine Learning Practice (0-1-1)	微控制器資訊感測 The sensing of microcontroller (0-1-1) 行動影音技術專題 Audiovisual Technology in Mobile Devices (0-1-1) iOS 行動應用實務 iOS Mobile App Practices (0-1-1)	機器手臂應用 Robot Arm Applications (0-1-1) CTF 資安實戰 CTF and Security (0-1-1) AI 生醫感測實務 AI Biomedical Sensor and Measurement (0-1-1)	感測器應用實務 Sensor Applications (0-1-1) 機器學習與資訊安全實務 Machine learning and Information Security Practice (0-1-1) 物聯網應用實務 IoT Appilication Practice (0-1-1)	專業養成課程 Professional Development Courses

跨域選修科目(合計 9 學分) Interdisciplinary Elective Courses (Total: 9 Credits)

*註：可選修外系專業選修，但不可為學術倫理、專題製作、專題研究、創新智慧生活科技、校外實習等課程。

*Note：Students may take professional electives from other departments, but the courses must not include Academic Ethics, Project Production, Research Projects, Innovative Smart Living Technology, or Off-Campus Internships.

最低畢業學分：130 學分 Minimum for Total Credits: 130 Credits

註：

1. **本系畢業學分為 130 學分，必修 103 學分、選修 27 學分、跨域選修 9 學分。**
   The total graduation requirement for this department is **130 credits**, including **103 required credits**, **27 elective credits**, and **9 interdisciplinary elective credits**.
2. 選修科目，本系得視實際需求修訂之。
   The elective courses may be revised by the department as needed.
3. 專業必修課程第一次修習於本系修課，再次修習時須事先徵得系之同意後可在他系修習同性質之課程。
   Professional required courses must be taken within this department for the first attempt. If retaken, approval from the department is required before enrolling in an equivalent course offered by another department.
4. **學生須修畢本校所開設之任一跨院系學分學程，始得畢業。**
   Students must complete at least one inter-college or interdisciplinary credit program offered by the university in order to graduate.
5. 上列**專業選修**科目，自行網路選課，低年級除特殊情況經系主任同意者外不得修習高年級課程。

The above professional elective courses should be selected online by the students themselves. Lower-year students are not allowed to take upper-year courses unless special permission is granted by the department chair.

6. **學生除選修本系專業選修課程外，可多選修產業學程、跨院系跨領域學程及課程等，以符合就業市場之需求，增加就業力。**
   In addition to taking the department's professional elective courses, students are encouraged to take additional industry-oriented, interdisciplinary, or cross-college programs and courses to meet job market demands and enhance their employability
7. 暑期校外實習、專業校外實習(一)課程皆於於暑期實施，前 160 小時為必修，可抵暑期校外實習 2 學分，第 161-320 小時為選修，可抵專業校外實習(一)2 學分；實習期滿成績及格者方給予承認課程並抵免學分。
   The Summer Off-Campus Internship and Professional Off-Campus Internship (I) courses are both conducted during the summer term. The first 160 hours are **required**, equivalent to **2 credits** for the Summer Off-Campus Internship. Hours **161–320** are **elective**, equivalent to **2 credits** for Professional Off-Campus Internship (I). Credits will be granted only upon successful completion of the internship with a **passing grade**.
8. 學期校外實習之實施為開設 9 學分以上，至少為期 4.5 個 月之校外實習課程，修讀實習課程期間，除依各校訂定定期返校之座談會或研習活動等外，學生應全職於實習機構實習。
   The semester-based off-campus internship is a course offering **9 or more credits**, requiring a minimum duration of **4.5 months**. During the internship period, except for scheduled on-campus seminars or workshops as stipulated by the university, students are expected to engage in **full-time internship work** at their assigned organizations.
9. **系專業必修科目「專業認證」，需於在本校就讀期間取得本系所認定之乙級以上（含）專業證照 1 張、取得參加全國性資訊相關競賽證明乙份、通過系專題展及繳交專題報告，始得通過。**
   For the departmental required course **"Professional Certification"**, students must, during their period of study at the university, obtain **at least one professional certificate at Class B or higher** as recognized by the department, **one proof of participation in a national information-related competition**, and **successfully complete the departmental project exhibition and submit a project report** in order to pass the course.